CSE 31
Computer Organization

Lecture 10 – C Memory Management

Announcements

- Lab
  - Lab 2 due this week (with 7 days grace period* after due date)
    - » Demo is REQUIRED to receive full credit**
  - Lab 3 out this week
    - » Due at 11:59pm on the same day of your next lab (with 7 days grace period* after due date)
    - » You must demo your submission to your TA within **14 days from posting of lab**
    - » Demo is REQUIRED to receive full credit**
- Reading and Homework assignments
  - Reading 01 (zyBooks 1.1 – 1.5) due 23-FEB
  - Homework 01 (zyBooks 1.1 – 1.5) due 02-MAR
    - » Complete Participation Activities in each section to receive grade towards Reading
    - » Complete Challenge Activities in each section to receive grade towards Homework
    - » IMPORTANT: Make sure to submit score to CatCourses by using the link provided on CatCourses

* There is a 10% penalty for late submissions during the grace period

** Submissions without demo will only receive partial credit for the TPS Activity portion of the assignment

Announcements

- Project 01
  - Due Fri, 20-MAR
  - Can work in teams of 2 students
    - » Each team member must identify teammate in "Comments…" text-box at the submission page
    - » If working in teams, each student must submit code (can be the same as teammate) and demo individually
    - » Grade can vary among teammates depending on demo
  - Demo required for project grade
    - » No partial credit for submission without demo
  - No grace period
    - » Must complete submission and demo by due date.

C Memory Management

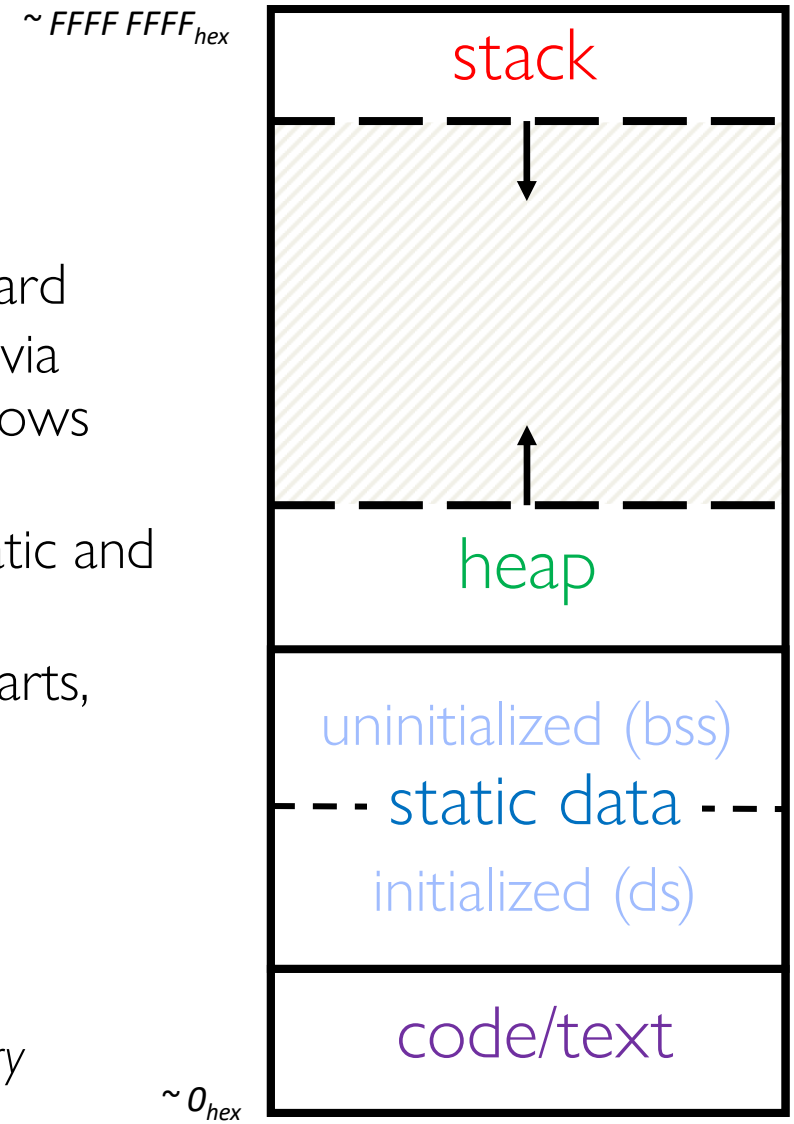
- C has 3 pools of memory (based on the nature of usage)
  - Static storage: global and static variable storage, basically permanent, entire program run
  - The Stack: local variable storage, parameters, return address (location of "activation records" in Java or "stack frame" in C)
  - The Heap (dynamic malloc storage): data lives until deallocated by programmer

- C requires knowing where things are in memory, otherwise things don't work as expected
  - Java hides location of objects

Normal C Memory Management

- A program's address space contains 4 regions:
  - stack: local variables, grows downward
  - heap: space requested for pointers via `malloc()`; resizes dynamically, grows upward
  - static data: Initialized/uninitialized static and global variables
  - code/text: loaded when program starts, does not change

For now, OS somehow prevents accesses between stack and heap (gray hash lines). Wait for virtual memory



Where are variables allocated?

- If declared outside of any function (global) or using the static keyword
  - allocated in "static" storage

- If declared inside of a function
  - allocated in the "stack" (unless declared as static)
  - freed when a function returns.
    - » That's why the scope is within the function

- Note: `main()` is a function!

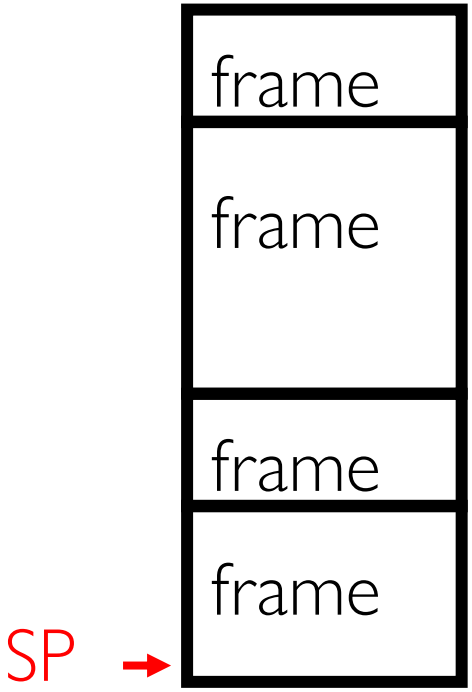
```
int myGlobal;
main() {
   static char myStatic;
   double myTemp;
}
```

Stack frames

- Stack frame includes storage for:
  - Return "instruction" address
  - Parameters (input arguments)
  - Space for other local variables

- Stack frames:
  - contiguous blocks of memory for a function
  - stack pointer tells where top stack frame is

- When a function ends, stack frame is "popped off" the stack; frees memory for future stack frames



Stack

- Last In, First Out (LIFO) data structure

stack

```
main (){
 a(0);
}
void a (int m) {
  b(1);
}
void b (int n){
 c(2);
}
void c (int o){
 d(3);
}
void d (int p) {
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Stack grows down

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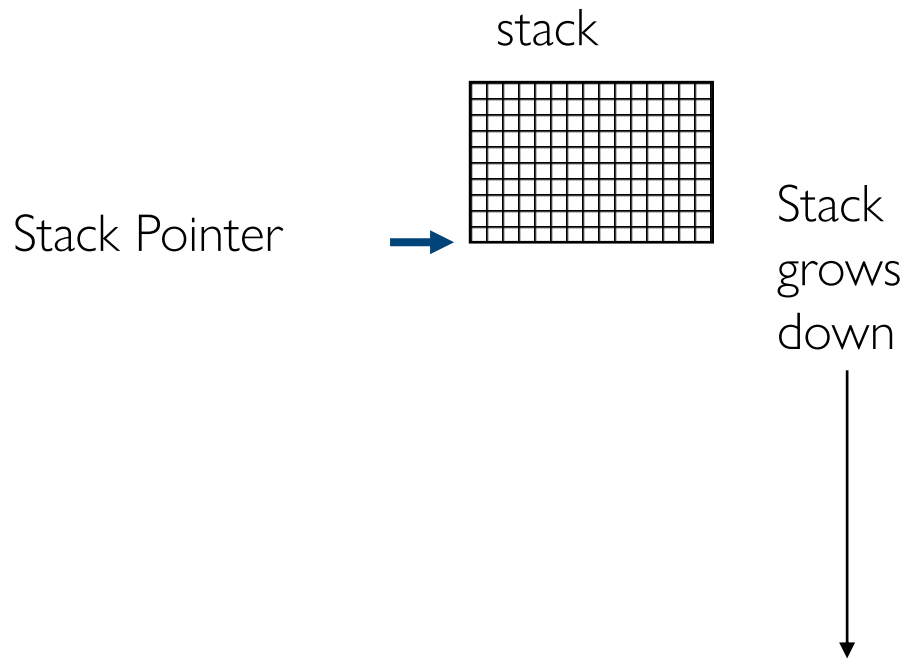

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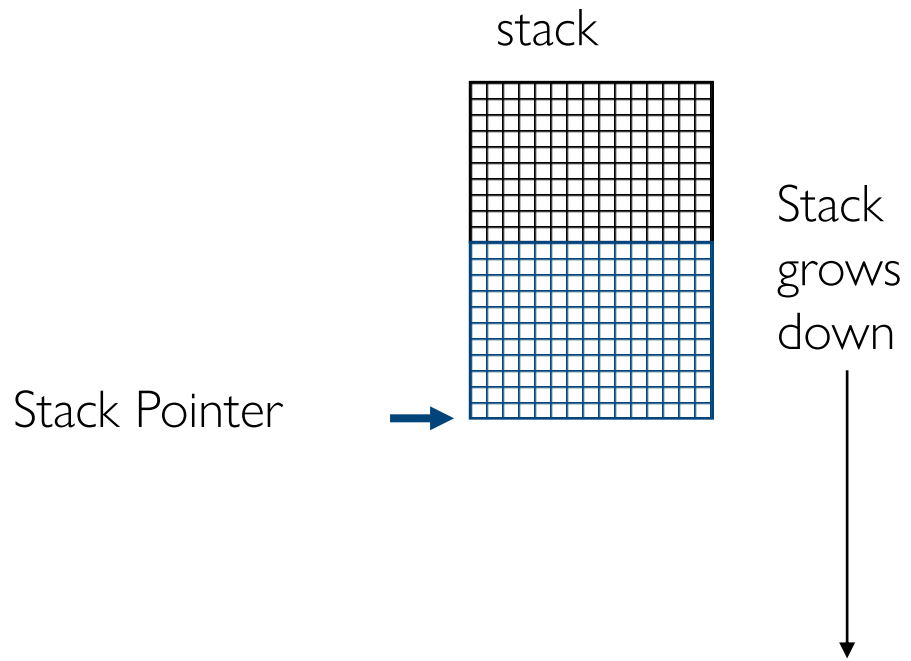

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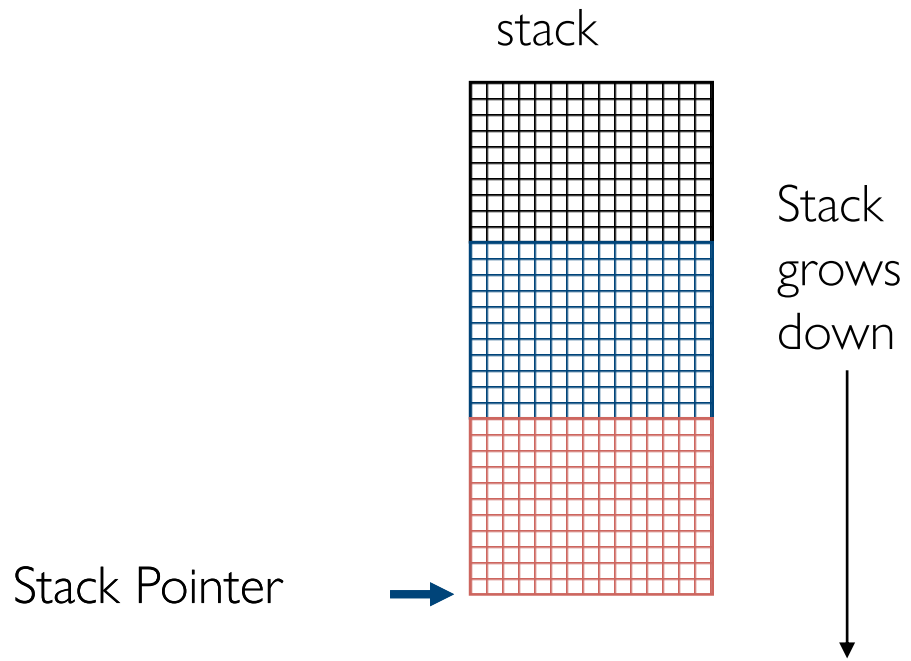

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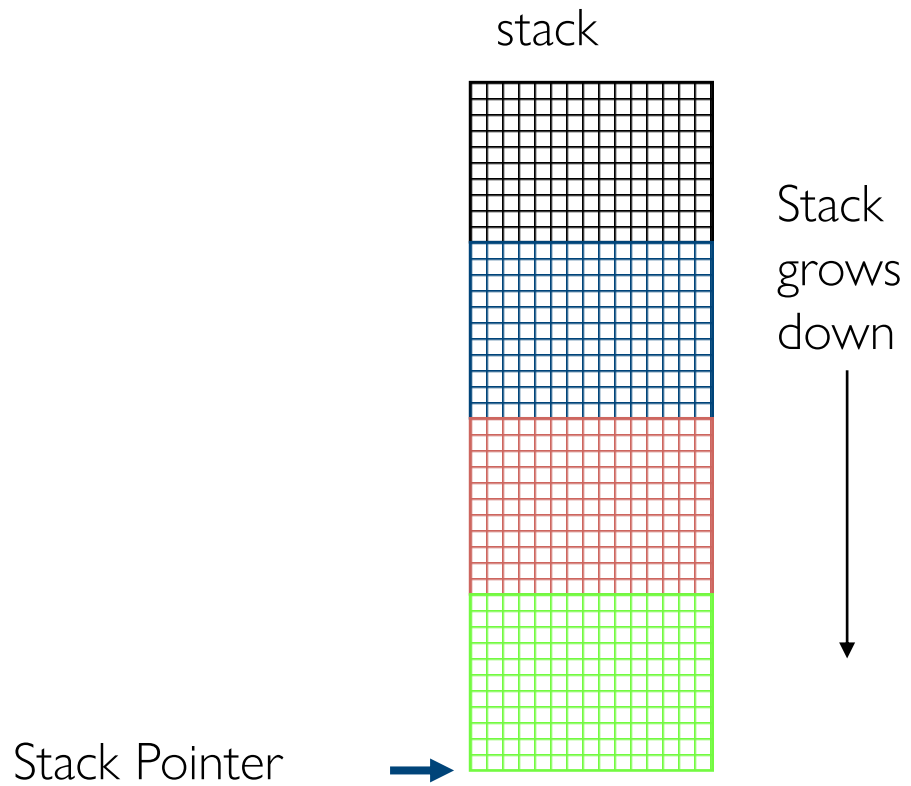

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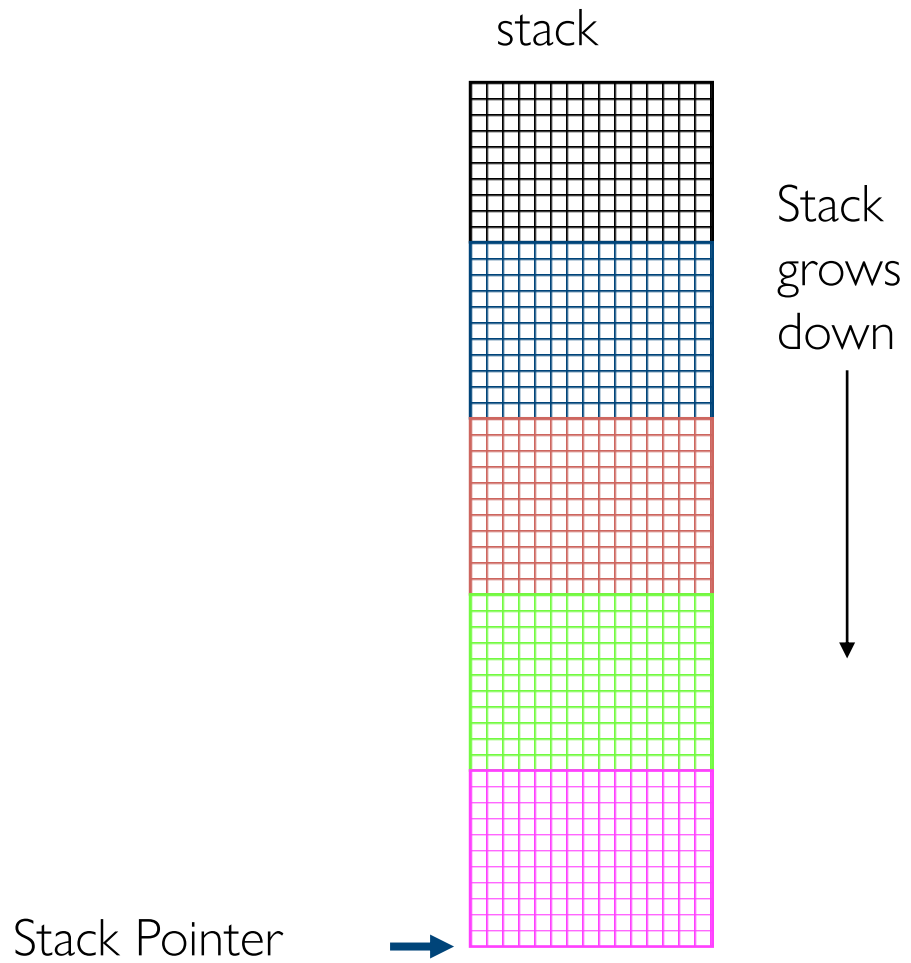

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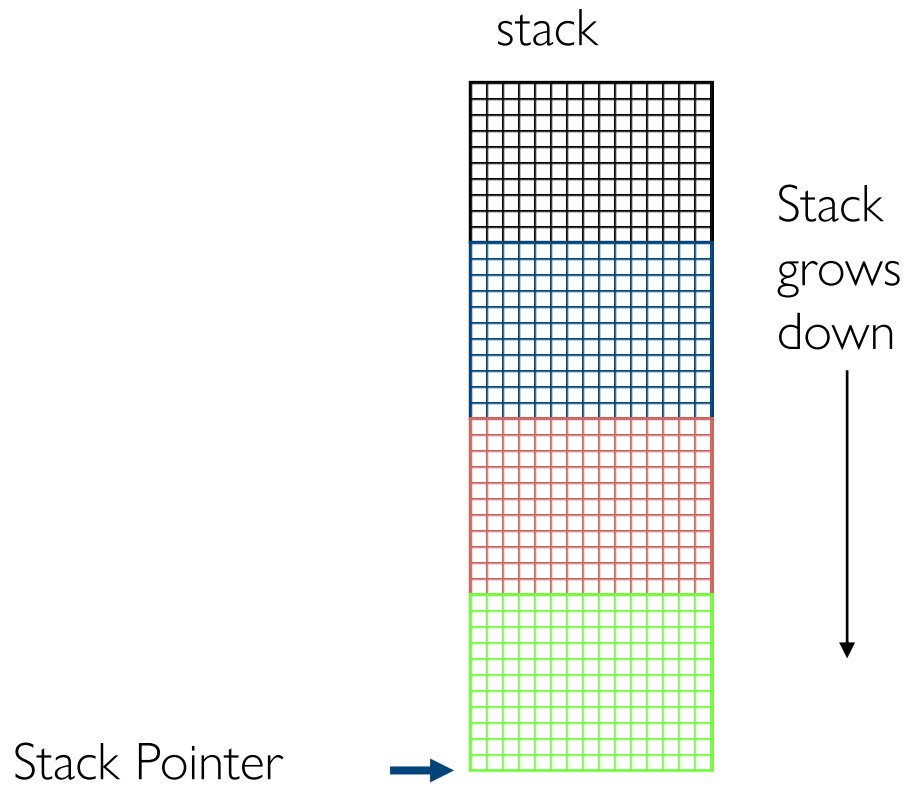

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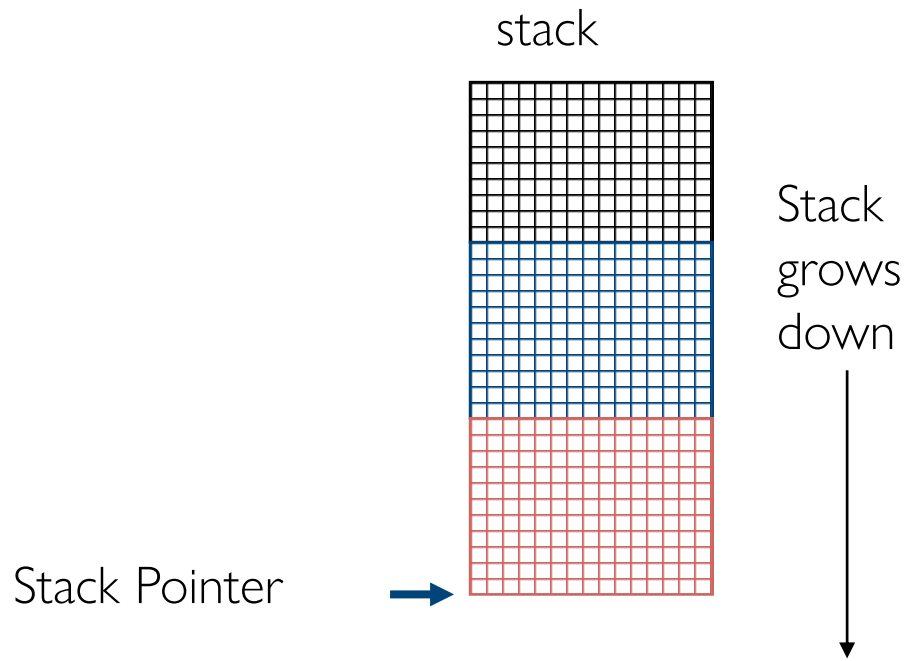

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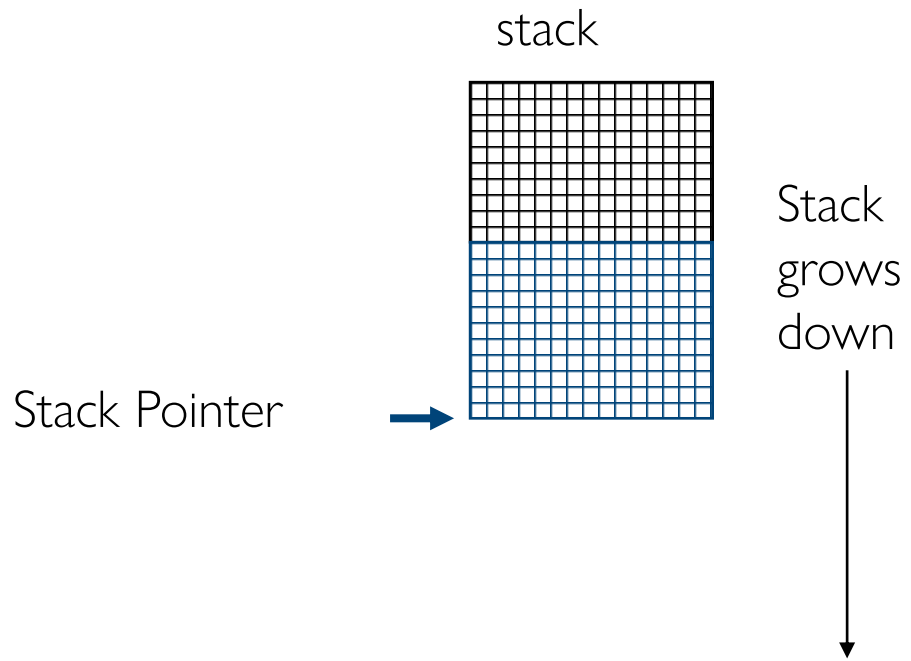

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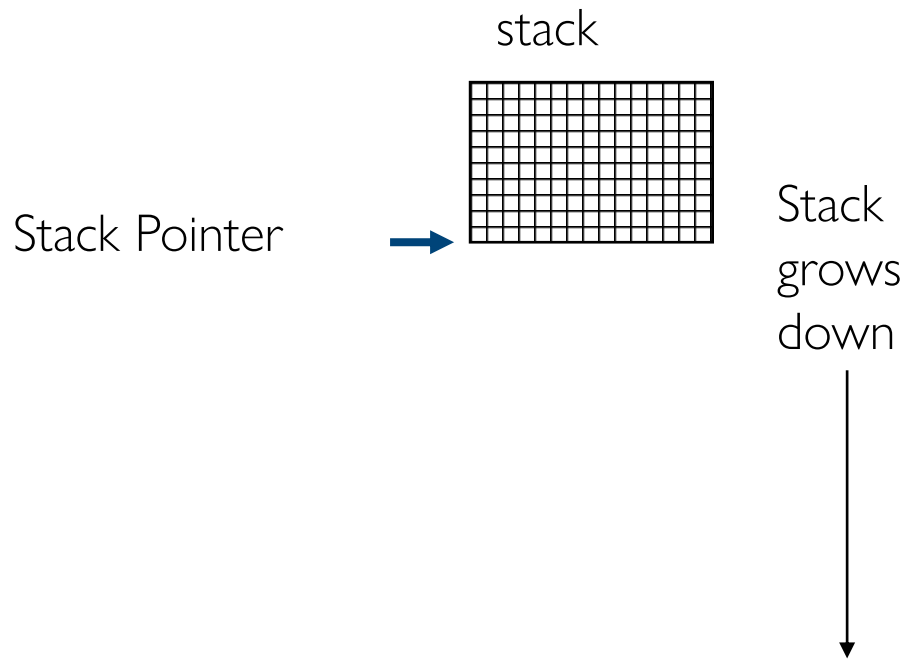

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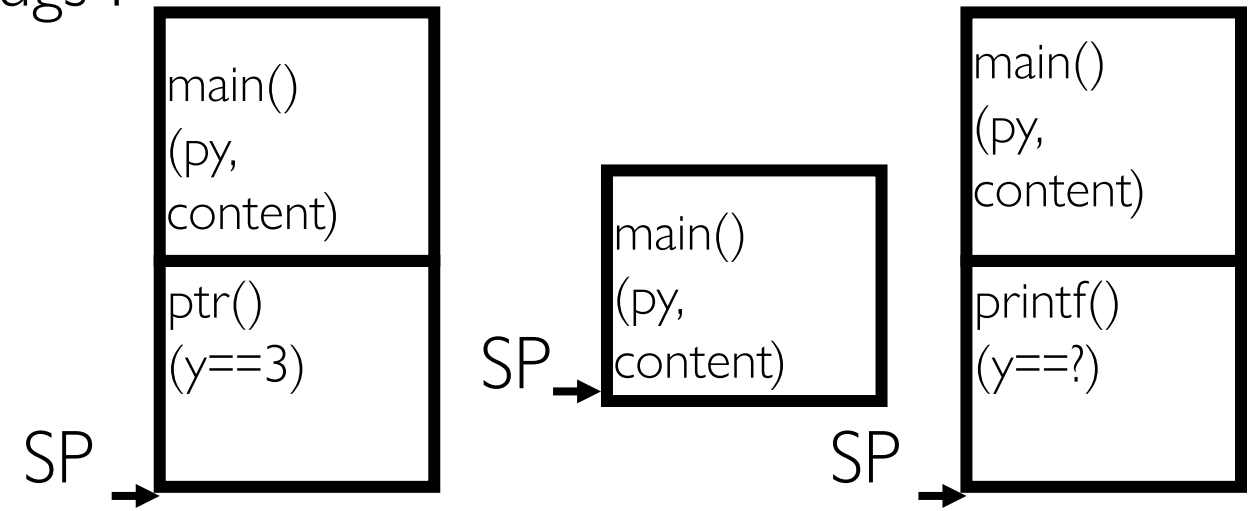
Stack grows down

Who cares about stack management?

- Pointers in C allow access to deallocated memory, leading to hard-to-find bugs !




```
int *ptr () {
    int y;
    y = 3;
    return &y;
}
```

```
int main () {
        int *py, content;
        py = ptr();
        content = *py;
        printf("%d", content); /* 3 */
        return 0;
}
```